SET 17

PART A — Short questions and answers

Q1. Summarize the symbols used for entities, attributes and relationships in an ER diagram.

Answer: This concept helps organize, retrieve, or maintain database information. Use the example named in the question to explain it in one or two sentences.

Q2. Interpret the statement: ALTER TABLE DISH ADD Is_Veg CHAR(1);

Answer: This concept helps organize, retrieve, or maintain database information. Use the example named in the question to explain it in one or two sentences.

Q3. Explain why storing Ingredients as a list in one column violates 1NF.

Answer: This concept helps organize, retrieve, or maintain database information. Use the example named in the question to explain it in one or two sentences.

Q4. Interpret the query: SELECT DishName FROM DISH WHERE Price > (SELECT AVG(Price) FROM DISH);

Answer: This concept helps organize, retrieve, or maintain database information. Use the example named in the question to explain it in one or two sentences.

Q5. Describe how GROUP BY with SUM computes the revenue of each cuisine.

Answer: GROUP BY collects rows with the same value so aggregate functions can calculate a result for each group.

Q6. Describe the purpose of CREATE OR REPLACE PROCEDURE.

Answer: PL/SQL adds procedural programming to SQL. A procedure is a reusable named block that can take parameters.

Q7. Describe how a one-to-many menu relationship can be represented with an array in a document.

Answer: This concept helps organize, retrieve, or maintain database information. Use the example named in the question to explain it in one or two sentences.

Q8. Explain how documents in the same collection can have different fields.

Answer: This concept helps organize, retrieve, or maintain database information. Use the example named in the question to explain it in one or two sentences.

Q9. Explain what the following MongoDB query returns: db.dishes.find({IsVeg: true, Price: {$lt: 200}})

Answer: The query returns all documents that satisfy the specified filter condition.

Q10. Explain why DETACH DELETE is needed to delete a node that has relationships.

Answer: This concept helps organize, retrieve, or maintain database information. Use the example named in the question to explain it in one or two sentences.

PART B — System solution

Scenario: A restaurant maintains a Restaurant Order Management System to store customer details, dish details,

1(a) ER design:

• **CUSTOMER**(CustomerID, CustomerName, Phone); primary key: **CustomerID**.

• **DISH**(DishID, DishName, Cuisine, Price); primary key: **DishID**.

• **FOOD_ORDER**(OrderID, CustomerID, DishID, OrderDate, Quantity, TableNo); primary key: **OrderID**.

Connect the two main entity tables through the transaction/relationship table. The transaction table has its own primary key and foreign keys to both entity tables. Cardinality: each entity can participate in many transactions; each transaction belongs to one row from each entity table (1:M on both sides, resolving M:N). Include the transaction's date/status/amount attributes shown in the question.

1(b) SQL DDL/DML: Create each table with its listed columns, PRIMARY KEY on its ID, and FOREIGN KEYs in the transaction table. Add NOT NULL to required fields; use CHECK/UNIQUE/DEFAULT where the question asks. Insert five sample rows per table, making sure foreign-key values exist.

```
CREATE TABLE CUSTOMER (
  CustomerID NUMBER PRIMARY KEY,
  CustomerName VARCHAR2(100),
  Phone VARCHAR2(100)
);
CREATE TABLE DISH (
  DishID NUMBER PRIMARY KEY,
  DishName VARCHAR2(100),
```

```
  Cuisine VARCHAR2(100),
  Price NUMBER
);
CREATE TABLE FOOD_ORDER (
  OrderID NUMBER PRIMARY KEY,
  CustomerID NUMBER,
  DishID NUMBER,
  OrderDate DATE,
  Quantity NUMBER,
  TableNo VARCHAR2(100)
);
```

2(a) Join query: Select the two entity names/details from the transaction table by joining the transaction table to both entity tables using their IDs.

```
SELECT a.*, b.*
FROM FOOD_ORDER x
JOIN CUSTOMER a ON x.CustomerID = a.CustomerID
JOIN DISH b ON x.DishID = b.DishID;
```

2(b) Aggregate query: Group the transaction rows by the requested category/type and apply HAVING to the aggregate condition stated in the question.

```
SELECT grouping_column, COUNT(*) AS total
FROM transaction_table
GROUP BY grouping_column
HAVING COUNT(*) > threshold;
```

2(c) PL/SQL procedure: Create a procedure with an IN parameter for the requested ID. Join the transaction table to the related entity tables, filter by that parameter, and print selected values using DBMS_OUTPUT.PUT_LINE.

```
CREATE OR REPLACE PROCEDURE show_details(p_id IN NUMBER) AS
BEGIN
  FOR r IN (
    SELECT a.*, b.*, x.*
    FROM transaction_table x
    JOIN entity_table_1 a ON x.entity1_id = a.entity1_id
    JOIN entity_table_2 b ON x.entity2_id = b.entity2_id
    WHERE x.transaction_id = p_id
  ) LOOP
    DBMS_OUTPUT.PUT_LINE('Details retrieved');
  END LOOP;
END;
/
```

3. MongoDB: Create the collections and insert documents using the fields listed in the set. Query with find() and the requested filter.

```
db.collection.insertMany([
  { _id: 1, name: 'Sample A', category: 'Example', price: 500 },
  { _id: 2, name: 'Sample B', category: 'Example', price: 300 }
]);
db.collection.find({ author: 'Specified Author' });
db.collection.find({ price: { $gt: 400 } });
```